

14.1 Measuring Temperature

Question Paper

Course	CIE A Level Physics
Section	14. Temperature
Topic	14.1 Measuring Temperature
Difficulty	Medium

Time allowed: 40

Score: /33

Percentage: /100

Question 1a

A metal plate is heated for a few minutes until it reaches thermal equilibrium.

State and explain what is meant by thermal equilibrium in this context.

[2 marks]

Question 1b

The temperature of the metal plate rises from $25\text{ }^{\circ}\text{C}$ to $450\text{ }^{\circ}\text{C}$.

On the thermodynamic scale of temperature, calculate

(i)
the temperature of $450\text{ }^{\circ}\text{C}$,

[2]

(ii)
the temperature rise of the block.

[1]

[3 marks]

Question 1c

Describe, in terms of particles, the transfer of energy as the metal plate reaches thermal equilibrium.

[2 marks]

Question 1d

A student states, incorrectly, that temperature measures the amount of thermal energy (heat) in a body.

State and explain **two** observations that show why this statement is incorrect.

[4 marks]

Question 2a

Explain, using thermal energy transfer, why blowing on a hot coffee cools it down faster.

[3 marks]

Question 2b

Fig 1.1 shows the cooling curve for a hot cup of coffee left standing in a room.

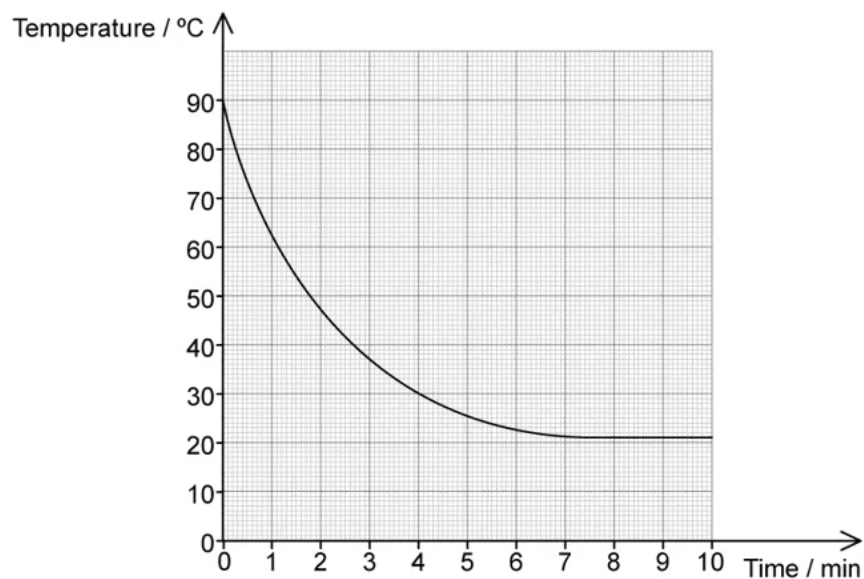


Fig 1.1

(i)

Determine the temperature of the room.

[1]

(ii)

Sketch on Fig 1.1 the curve for when the coffee is blown on.

[2]

[3 marks]

Question 2c

A student states that the temperature of the coffee measures the amount of thermal energy (heat) in it.

Suggest whether this statement is correct. Explain your reasoning using two observations.

[5 marks]

Question 3a

(i)

Suggest two instruments that can be used to measure the temperature of a water bath.

[2]

(ii)

State two reasons why they may record different temperatures.

[2]

[4 marks]

Question 3b

The e.m.f generated in a thermocouple thermometer may be used for the measurement of temperature.

Fig 1.1 shows the variation with temperature T of the e.m.f E .

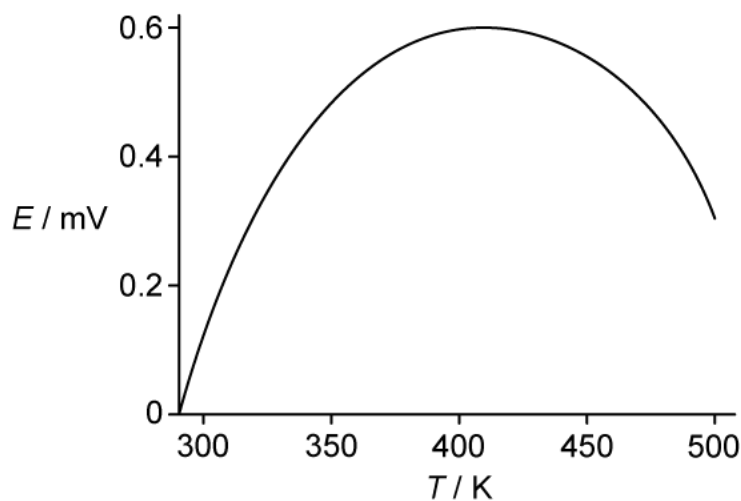


Fig 1.1

By reference to Fig 1.1, explain why this thermocouple would not be suitable to measure temperature.

[3 marks]

Question 3c

State two advantages and two disadvantages of using a resistance thermometer over a thermocouple thermometer to measure the temperature of an object.

[4 marks]

Model Answers: Medium

1a

(a) Thermal equilibrium means:

- The temperature of the metal plate is the same / equal at all points; [1 mark]
- There is no (net) transfer of energy (within the metal plate); [1 mark]

[Total: 2 marks]

An answer such as 'the temperature is the same at all points' will not get you the mark, as it is too vague. You must state the temperature of **what** is the same at all points, since the questions as asked you to refer to the **context**.

1b

(b)

(i) Convert 450 °C into Kelvin:

- Thermodynamic temperature: $T / K = \theta / ^\circ\text{C} + 273.15$
- $T / K = \theta / ^\circ\text{C} + 273$ [1 mark]
- $T / K = 450 + 273 = 723 \text{ K}$ [1 mark]

(ii) Determine the temperature rise in Kelvin:

- Change in temperature = $450 - 25 = 425 \text{ K}$ [1 mark]

[Total: 3 marks]

Remember - when you calculate a temperature **change** in Kelvin you don't need to add 273! The **difference** between two points on the Celsius or Kelvin scale is the same.

1c

(c) As the metal plate reaches thermal equilibrium:

Any **two** from:

- Energy is transferred through the metal as electrons move; [1 mark]
- In the region of high temperature, the particles vibrate more **OR** in the region of low temperature, the particles vibrate less; [1 mark]
- The vibrations are passed along the lattice / neighbouring particles; [1 mark]
- From (the region of) higher temperature to (the region of) lower temperature until the energy has been distributed evenly; [1 mark]

[Total: 2 marks]

1d

(d) Observations that show why this statement is incorrect are:

Any **two** pairs from:

- Two bodies of different masses can have the same temperature; [1 mark]
- But a different amount of heat (energy); [1 mark]
- Temperature shows the direction of heat transfer; [1 mark]
- From a region of high temperature to low temperature; [1 mark]
- When a substance changes state (boils / melts etc); [1 mark]
- Heat energy can be put in, but there is no temperature change; [1 mark]

[Total: 4 marks]

2a

(a) Blowing on a hot coffee cools it down faster because:

- The air that comes in to contact with the coffee is cooler than the coffee / liquid (itself); [1 mark]
- Heat / thermal energy transfers from regions of higher temperature (the coffee) to lower temperature (the surrounding air); [1 mark]
- Therefore, (the coffee) reaches thermal equilibrium (with the surroundings) faster; [1 mark]

[Total: 3 marks]

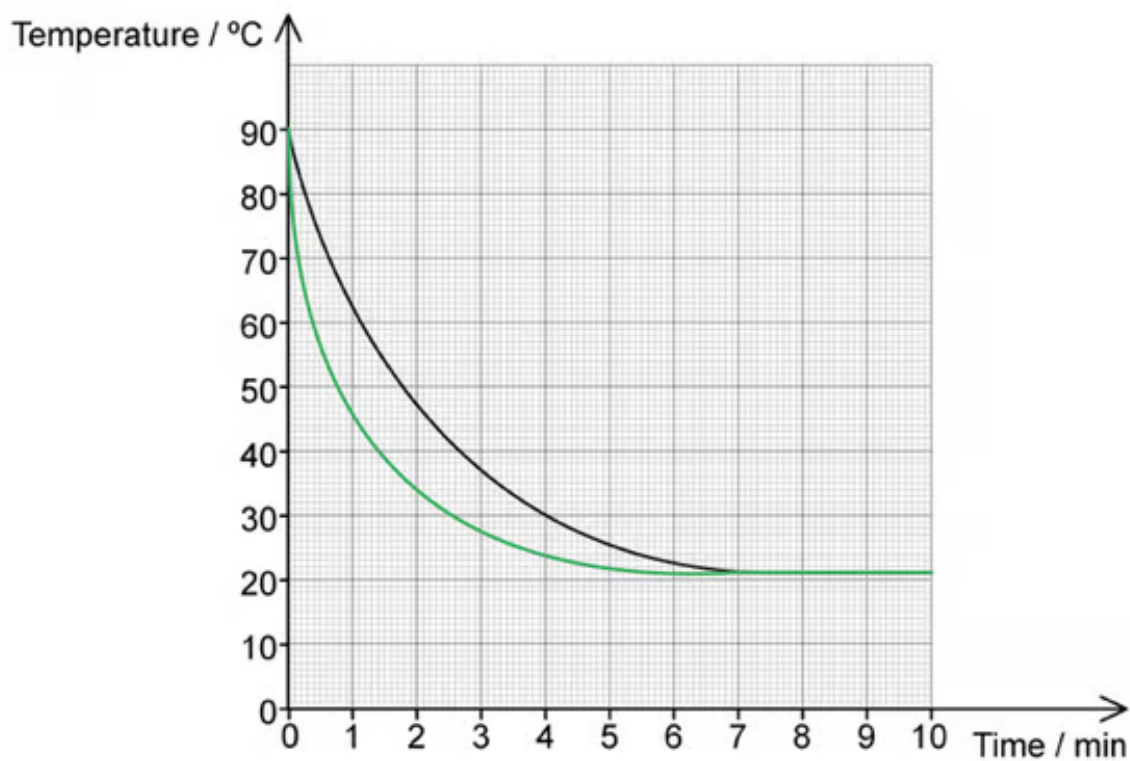
2b

(b)

(i) The temperature of the room is:

- 21°C [1 mark]

(ii) The curve for when the coffee is blown on should look like:



- Curve steeper at the start; [1 mark]
- Still plateauing at 21°C [1 mark]

[Total: 3 marks]

When sketching the graph in part (ii), make sure it still plateaus at room temperature (i.e. the temperature of the coffee has reached thermal equilibrium with the temperature of the room). Blowing on it only changes the **rate** of cooling, not its final temperature, which can only be changed if the temperature of the room changes.

2c

(c) This statement is:

- Incorrect; [1 mark]

Because:

Any **two** pairs from:

- For two objects of different masses at the same temperature; [1 mark]
- The same material would have a different amount of heat; [1 mark]
- Temperature shows the direction of the heat transfer; [1 mark]
- From high to low regardless of the objects; [1 mark]
- When a substance melts / boils; [1 mark]
- The heat input causes no temperature change; [1 mark]

[Total: 5 marks]

3a

(a)

(i) Two instruments that can be used to measure the temperature of a water bath are:

- A resistance thermometer; [1 mark]
- A thermocouple; [1 mark]

(ii) They may record different temperatures because:

Any **two** from:

- They have not been calibrated; [1 mark]
- They are not in (thermal) equilibrium with the water in the bath; [1 mark]

Because:

Any **two** pairs from:

- For two objects of different masses at the same temperature; [1 mark]
- The same material would have a different amount of heat; [1 mark]
- Temperature shows the direction of the heat transfer; [1 mark]
- From high to low regardless of the objects; [1 mark]
- When a substance melts / boils; [1 mark]
- The heat input causes no temperature change; [1 mark]

[Total: 5 marks]

3a

(a)

(i) Two instruments that can be used to measure the temperature of a water bath are:

- A resistance thermometer; [1 mark]
- A thermocouple; [1 mark]

(ii) They may record different temperatures because:

Any **two** from:

- They have not been calibrated; [1 mark]
- They are not in (thermal) equilibrium with the water in the bath; [1 mark]
- A calibrated temperature scale assumes a linear change of property with temperature; [1 mark]
- Neither property varies linearly with temperature; [1 mark]

[Total: 4 marks]

3b

(b) This thermocouple would not be suitable to measure temperature because:

- After (an e.m.f. of) 0.4 mV; [1 mark]
- The variation is non-linear / curved; [1 mark]
- So there are two possible temperatures; [1 mark]

[Total: 3 marks]

3c

(c) Two advantages and two disadvantages of using a resistance thermometer over a thermocouple thermometer to measure the temperature of an object are:

Two advantages:

Any **two** from:

- It has a smaller thermal capacity; [1 mark]
- It can measure $\Delta\theta$ of a small object; [1 mark]
- It is more sensitive; [1 mark]
- The readings are taken at a point / are smaller; [1 mark]
- It can be used to measure the temperature difference; [1 mark]
- No power supply required; [1 mark]
- The readings are more accurate / it has more accuracy; [1 mark]

Two disadvantages:

Any **two** from:

- It is more expensive / a thermocouple is cheaper; [1 mark]
- It can measure a smaller range of temperatures (than a thermocouple); [1 mark]
- It has a slower reaction time; [1 mark]

[Total: 4 marks]

